Real Analysis Seminar.

I. Measure Theory on $\mathbb{R}^d$

1. Exterior measure.

Def I.1.1. $E \subseteq \mathbb{R}^d$, then the exterior measure of E, denoted by $m_*(E) := \inf\left\{\sum_{j=1}^{\infty}|Q_j| \,\middle|\, E \subseteq \bigcup_{j=1}^{\infty} Q_j\right\}$ where Q_j $(j \geq 1)$ are closed cubes.

Lemma 0. (1) Let $E \subseteq \mathbb{R}^d$ be a rectangle, $E \subseteq \bigcup_{j=1}^{N} Q_j$. Q_j are cubes. then $|E| \leq \sum_{j=1}^{N}|Q_j|$

(2) Every open subsets O of $\mathbb{R}^d$ can be written as a countable union of almost disjoint closed cubes

pf: (skipped)

Prop. I.1.2. (1) If $E_1 \subseteq E_2$. then $m_*(E_1) \leq m_*(E_2)$.

(2) If $E \subseteq \bigcup_{j=1}^{\infty} E_j$. then $m_*(E) \leq \sum_{j=1}^{\infty} m_*(E_j)$

(3) If $E \subseteq \mathbb{R}^d$ then $m_*(E) = \inf\{m_*(O) \mid E \subseteq O \text{ open}\}$.

(4) If $E = E_1 \cup E_2$. and $d(E_1, E_2) > 0$. then $m_*(E) = m_*(E_1) + m_*(E_2)$

(5) If $E = \bigcup_{j=1}^{\infty} Q_j$. Q_j are almost disjoint cubes. then $m_*(E) = \sum_{j=1}^{\infty} m_*(Q_j)$.

Proof: We just show (4): First, it's clear to have $m_*(E) \leq m_*(E_1) + m_*(E_2)$.

$\forall \varepsilon > 0$, Now suppose $d(E_1, E_2) \geq \delta > 0$. let $E \subseteq \bigcup_{j=1}^{\infty} Q_j$ closed cubes s.t. $m_*(E) + \varepsilon \geq \sum_{j=1}^{\infty}|Q_j|$.

Divide Q_j such that diam $Q_j < \delta \Rightarrow$ Then either $Q_j' \cap E_1 = \phi$ or $Q_j' \cap E_2 = \phi$.

Then we have $E_1 \subseteq \bigcup_{j \in J_1} Q_j$. $E_2 \subseteq \bigcup_{j \in J_2} Q_j$. $(J_1 \cup J_2 = \mathbb{N}_+,\ J_1 \cap J_2 = \phi)$.

Thus $m_*(E_1) + m_*(E_2) \leq \sum_{j \in J_1}|Q_j| + \sum_{j \in J_2}|Q_j| = \sum_{j=1}^{\infty}|Q_j| \leq m_*(E) + \varepsilon$ □.

2. Measurabe sets and the Lebsgue measure.

Def I.2.1. $E \subseteq \mathbb{R}^d$ is Lebesgue measurable if $\forall \varepsilon > 0$. $\exists$ open set O such that $E \subseteq O$. and $m_*(O - E) < \varepsilon$. And the measure of E, $m(E) := m_*(E)$.

Prop. I.2.2. (1) Every open subset $\subseteq \mathbb{R}^d$ is measurable

(2) If $m_*(E) = 0$. then E is measurable

(3) If E_j is measurable, then $\bigcup_{j=1}^{\infty} E_j$ is measurable.

(4) Closed sets are measurable

(5) If E is measurable. then $\mathbb{R}^d \setminus E$ is measurable

(6) If E_j is measurable, then $\bigcap_{j=1}^{\infty} E_j$ is measurable.

proof: (1) trivial;

(2) $m_*(E) = \inf\{m_*(O) \mid E \subseteq O \text{ open}\} = 0$. $\forall \varepsilon > 0$, $\exists O \supseteq E$ open s.t. $m_*(O) < \varepsilon$

$\Rightarrow m_*(O - E) \leq m_*(O) < \varepsilon$

(3) $\forall \varepsilon > 0, \exists O_j \supseteq E_j$ open. s.t. $m_*(O_j - E_j) < \frac{\varepsilon}{2^{j+1}}$ $\Rightarrow m_*\left(\bigcup_{j\geq 1} O_j - \bigcup_{j\geq 1} E_j\right)$

$\leq \sum_{j=1}^{\infty} m_*(O_j - E_j) < \varepsilon$ □

(4) $F = \bigcup_{j\geq 1}(F \cap B_k)$, $B_k = \overline{B(0,k)}$ then by (3) It suffices to show the situation of compact sets. Assume that F is compact

$\exists O \supseteq F$ s.t. $m_*(O) + \varepsilon \geq m_*(F)$.

Since $O - F$ is open, by lemma we have $O - F = \bigcup_{j\geq 1} Q_j$ almost disjoint cubes

$K = \bigcup_{j=1}^{N} Q_j$ is compact $\Rightarrow d(F, K) > 0$, then $m_*(F \cup K) = m_*(F) + m_*(K)$

$\Rightarrow m_*(O) \geq m_*(F) + m_*(K) = m_*(F) + \sum_{j=1}^{N} |Q_j| \ (\forall N \geq 1)$.

$\Rightarrow m_*(O - F) \leq \sum_{j=1}^{\infty} |Q_j| \leq m_*(O) - m_*(F) < \varepsilon$ □.

(5). E is measurable. $\Rightarrow \exists O_n \supseteq E$, s.t. $m_*(O_n - E) < \frac{1}{n}$. (open)

let $S = \bigcup_{n\geq 1} O_n^c$ then S is measurable. $\Rightarrow m_*(E^c - S) \leq m_*(O_n - E) < \frac{1}{n}$

$\Rightarrow m_*(E^c - S) = 0$ thus measurable,

(6) $\left(\bigcup_{j=1}^{n} E_j^c\right)^c = \bigcap_{j=1}^{n} E_j$ is measurable □.

Thm I.2.3. If E_j are disjoint measurable sets, then $m\left(\bigcup_{j=1}^{\infty} E_j\right) = m(E) = \sum_{j=1}^{\infty} m(E_j)$

proof: 1). E_j is bounded. $\exists F_j \subseteq E_j$ s.t. $m_*(E_j - F_j) < \frac{\varepsilon}{2^j}$

$\forall N > 0$. $m\left(\bigcup_{j=1}^{N} F_j\right) = \sum_{j=1}^{N} m(F_j)$ (F_j is compact and disjoint).

$\Rightarrow \forall N$, $m(E) \geq \sum_{j=1}^{N} m(F_j) \geq \sum_{j=1}^{N} m(E_j) - \varepsilon$ $\left(m(E_j - F_j) \geq m(E_j) - m(F_j)\right)$

$\Rightarrow m(E) \geq \sum_{j=1}^{\infty} m(E_j) - \varepsilon \Rightarrow m(E) \geq \sum_{j=1}^{\infty} m(E_j) \Rightarrow m(E) = \sum_{j\geq 1} m(E_j)$

2) let $\{Q_k\}_{k\geq 1}$ satisfy $Q_k \subseteq Q_{k+1}$, s.t. $\bigcup_{k\geq 1} Q_k = \mathbb{R}^d$. Q_k cubes

$S_k = Q_{k+1} - Q_k$, then $E = \bigcup_{j\geq 1}\bigcup_{k\geq 1}(S_k \cap E_j) := \bigcup_{j\geq 1}\bigcup_{k\geq 1} E_{j,k}$

$E_{j,k}$ is bounded $\Rightarrow m(E_j) = \sum_{k\geq 1} m(E_{j,k})$

$\Rightarrow m(E) = \sum_{j\geq 1}\sum_{k\geq 1} E_{j,k} = \sum_{j\geq 1} m(E_j)$ □

Def I.2.4. (i) $\{E_k\}_{k\geq 1}$ are sequences of subsets in $\mathbb{R}^d$ s.t. $E_k \subseteq E_{k+1}$, $E = \bigcup_{k=1}^{\infty} E_k$. then we write $E_k \nearrow E$.

(ii) $\{E_k\}_{k\geq 1}$, $E_k \supseteq E_{k+1}$ and $\bigcap_{k=1}^{\infty} E_k = E$. then we write $E_k \searrow E$.

Coro I.2.5. Suppose E_j are measurable

(i) If $E_k \nearrow E$. then $m(E) = \lim_{N\to\infty} m(E_N)$

(ii) If $E_k \searrow E$, $m(E_k) < \infty$ for some k, then $m(E) = \lim_{N\to\infty} m(E_N)$

(i) let $G_1 = E_1$. $G_k = E_k - E_{k-1}$. then $m(E) = \sum_{j\geq 1} m(G_j) = \lim_{N\to\infty} m(\bigcup_{j=1}^{N} G_j)$

$= \lim_{N\to\infty} m(E_N)$

(ii). Assume that $m(E_1) < \infty$. then. $G_k = E_k - E_{k+1} \Rightarrow E_1 = E \cup \bigcup_{k\geq 1} G_k$.

$\Rightarrow m(E_1) = m(E) + \lim_{N\to\infty} \sum_{k=1}^{N-1} (m(E_k) - m(E_{k+1})) = m(E) + m(E_1) - \lim_{N\to\infty} m(E_N)$

$\Rightarrow m(E) = \lim_{N\to\infty} m(E_N)$ □.

Thm I.2.6. Suppose E is a measurable subset of $\mathbb{R}^d$. Then $\forall \varepsilon > 0$,

(i). $\exists$ open $\mathcal{O}$ s.t. $E \subseteq \mathcal{O}$, $m(\mathcal{O} - E) < \varepsilon$.

(ii) $\exists$ close F s.t. $F \subseteq E$, $m(E - F) < \varepsilon$.

(iii) If $m(E) < \infty$, then $\exists$ compact $K \subseteq E$ s.t. $m(E - K) < \varepsilon$.

(iv) If $m(E) < \infty$, then $\exists$ Q_j close cubes s.t. $F = \bigcup_{j=1}^{N} Q_j$. $m(E \Delta F) < \varepsilon$.

proof: We just show (iii): $\exists$ close F s.t. $F \subseteq E$. $m(E - F) < \frac{\varepsilon}{2}$.

then let $B_k = \overline{B(0,k)}$. then $F = \bigcup_{k\geq 1} (F \cap B_k) := \bigcup_{k\geq 1} F_k$.

where F_k is compact. and $(E - F_k) \searrow (E - F)$ ($m(E) < \infty$)

$\Rightarrow \exists N > 0$ s.t. $m(E - F_N) < m(E - F) + \frac{\varepsilon}{2} < \varepsilon$ □

Coro I.2.7. A subset E of $\mathbb{R}^d$ is measurable

(i) iff E differs from a G_δ by a set of measure zero.

(ii) iff E differs from a F_σ by a set of measure zero.

proof: Only $\Rightarrow$: $E \subseteq \mathbb{R}^d$ is measurable.

(i) $\exists \mathcal{O}_n \supseteq E$ open s.t. $m(\mathcal{O}_n - E) < \frac{1}{n}$. Let $S = \bigcap_{n\geq 1} \mathcal{O}_n \supseteq E$

Then $m_*(S - E) \leq m_*(\mathcal{O}_n - E) < \frac{1}{n}$ $(\forall n) \Rightarrow m(S - E) = m_*(S - E) = 0$

(ii). $\exists F_n \subseteq E$ close s.t. $m(E - F_n) < \frac{1}{n}$ let $F_\sigma = \bigcup_{n\geq 1} F_n \subseteq E$.

Then $m_*(E - F_\sigma) \leq m_*(E - F_n) < \frac{1}{n}$ $(\forall n) \Rightarrow m(E - F_\sigma) = 0$ □

3. Measurable functions

Def I.3.1. (1) $E \subseteq \mathbb{R}^d$. A characteristic function of E, $\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$.

(2) A simple function, $f = \sum_{k=1}^{N} a_k \chi_{E_k}$ where a_k are constants., $m(E_k) < \infty$

(3) f is finite valued if $-\infty < f(x) < +\infty$ for all x.

(4) f defined on a measurable set $E \subseteq \mathbb{R}^d$ is measurable iff $\forall a \in \mathbb{R}$, the set $f^{-1}([-\infty, a)) = \{x \in E \mid f(x) < a\}$ is measurable.

Prop. I.3.2. (1) f is finite-valued, then f is measurable iff $f^{-1}(O)$ is measurable iff $f^{-1}(F)$ is measurable. where O open. F close.

Rmk. In the following chapters. we will extend $\mathbb{R}$ to $\overline{\mathbb{R}} := \mathbb{R} \cup \{+\infty\} \cup \{-\infty\}$ where the topology is generated by (a,b) $a<b \in \mathbb{R}$, $[-\infty, a)$, $(a, +\infty]$

It's easy to check that $\overline{\mathbb{R}}$ is a compact Hausdorff space. (But we'll not talk it further!)

(2) If f is continuous on $\mathbb{R}^d$ then f is measurable.

If f is measurable and finite-valued, and Φ is continuous, then $\Phi \circ f$ is continuous

(3) Suppose $\{f_n\}_{n \geq 1}$ are measurable. Then $\sup_{n \geq 1} f_n(x)$, $\inf_{n \geq 1} f_n(x)$, $\limsup_{n \to \infty} f_n(x)$, $\liminf_{n \to \infty} f_n(x)$ is measurable.

proof: We just show (3). ~~$\sup_n f_n$~~ $\{\sup_{n \geq 1} f_n > a\} = \bigcup_{n \geq 1} \{f_n > a\}$ is measurable.

$\{\limsup_{n \to \infty} f_n > a\} = \{\inf_k \{\sup_{n \geq k} f_n\} > a\} = \bigcap_{k \geq 1} \bigcup_{n \geq k} \{f_n > a\}$ is measurable □.

Prop. I.3.3. If $\{f_n\}_{n \geq 1}$ measurable, and $\lim_{n \to \infty} f_n(x) = f(x)$. then f is measurable.

pf: (sketch) $\lim_{n \to \infty} f_n = \limsup_{n \to \infty} f_n = \liminf_{n \to \infty} f_n$ □.

Prop. I.3.4. If f, g are measurable, then

(i) The integer powers f^k, $k \geq 1$ are measurable

(ii) If f, g are finite-valued, then $f+g$, fg are measurable.

(i) $2 \mid k \Rightarrow \{f^k > a\} = \{f > a^{\frac{1}{k}}\} \cup \{f < -a^{\frac{1}{k}}\}$.

$2 \nmid k \Rightarrow \{f^k > a\} = \{f > a^{\frac{1}{k}}\}$

(ii) $\{f+g > a\} = \bigcup_{r \in \mathbb{Q}} \{f > a-r\} \cap \{g > r\}$.

$fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$ thus fg is measurable □.

Def I.3.5. f and g defined on E are equal almost everywhere, $f(x) = g(x)$ a.e. $x \in E$ if $\{x \in E \mid f(x) \neq g(x)\}$ has measure zero.

Rmk. The above properties can be replaced by the conditions holding almost everywhere.

Thm I.3.6. f is a non-negative measurable function on $\mathbb{R}^d$. Then $\exists$ $\{\varphi_k\}_{k\geq 1}$ s.t. (φ_k simple)

$\varphi_k(x) \leq \varphi_{k+1}(x)$. $\lim_{k\to\infty} \varphi_k(x) = f(x)$ for all x.

Proof. let Q_{2^k} denote the cube with the side 2^k, centered at the origin.

$$F_{2^k}(x) = \begin{cases} f(x) & x \in Q_{2^k} \wedge f(x) \leq 2^k \\ 2^k & x \in Q_{2^k} \wedge f(x) > 2^k \\ 0 & \text{otherwise.} \end{cases}$$

Then $\lim_{k\to\infty} F_{2^k}(x) = f(x)$.

Let $E_{l,2^k} = \{x \in Q_{2^k} \mid \frac{l}{2^k} < F_{2^k}(x) \leq \frac{l+1}{2^k}\}$. $0 \leq l < 2^{2k}$

$\varphi_k(x) = F_{2^k, 2^k}(x) = \sum_{l=1}^{2^{2k}} \frac{l}{2^k} \chi_{E_{l,2^k}}(x)$ $\Rightarrow \varphi_k(x) \leq \varphi_{k+1}(x)$

$\Rightarrow 0 \leq F_{2^k}(x) - F_{2^k,2^k}(x) \leq \frac{1}{2^k}$ for all x. Then we construct the desired sequence $\square$

Thm I.3.7. f is measurable on $\mathbb{R}^d$. Then $\exists$ simple $\{\varphi_k\}_{k\geq 1}$ s.t.

$|\varphi_k(x)| \leq |\varphi_{k+1}(x)|$, $\lim_{k\to\infty} \varphi_k(x) = f(x)$ for all x.

Proof: let $f(x) = f^+(x) - f^-(x)$, $f^+(x) = \max\{f(x), 0\}$, $f^-(x) = \max\{-f(x), 0\}$ $\square$

Thm. I.3.8. f is measurable on $\mathbb{R}^d$. Then $\exists$ step functions $\{\psi_k\}_{k\geq 1}$ s.t. $\psi_k \to f$ a.e. x.

Proof: (skipped)

4. Littlewood's three principles:

(i) Every set is nearly a finite union of intervals

(ii) Every function is nearly continuous

(iii) Every convergent sequence is nearly uniformly convergent.

Thm I.4.1. (Egorov) $\{f_k\}_{k\geq 1}$ measurable defined on a measurable set E, $m(E)<\infty$. Assume that $f_k \to f$ a.e. on E. Given $\varepsilon>0$ $\exists A_\varepsilon$ closed s.t. $A_\varepsilon \subseteq E$. $m(E-A_\varepsilon)<\varepsilon$. and $f_k \to f$ uniformly on A_ε.

Proof: W.L.O.G $f_k(x) \to f(x)$, $\forall x \in E$; let $E_k^n = \{x \in E \mid |f_j(x)-f(x)|<\frac{1}{n}$, for all $j>k\}$.

Then $E_k^n \subseteq E_{k+1}^n$, $E_k^n \nearrow E$ when $k \to \infty$

$\exists k_n$. s.t. $m(E-E_{k_n}^n)<\frac{1}{2^n}$

Then $|f_j(x)-f(x)|<\frac{1}{n}$, when $j>k_n$ and $x \in E_{k_n}^n$

Choose N s.t. $\sum_{n=N}^{\infty} 2^{-n} < \frac{\varepsilon}{2}$, let $\tilde{A}_\varepsilon = \bigcap_{n\geq N} E_{k_n}^n$

$\Rightarrow m(E-\tilde{A}_\varepsilon) \leq \sum_{n=N}^{\infty} m(E-E_{k_n}^n) < \frac{\varepsilon}{2}$.

$\forall \delta>0$. choose $n \geq N$. s.t. $\frac{1}{n}<\delta \Rightarrow \forall x \in \tilde{A}_\varepsilon$, $x \in E_{k_n}^n \Rightarrow |f_j(x)-f(x)|<\frac{1}{n}<\delta$. $\forall j>k_n$

Choose $A_\varepsilon \subseteq \tilde{A}_\varepsilon$ closed s.t. $m(\tilde{A}_\varepsilon - A_\varepsilon)<\frac{\varepsilon}{2} \Rightarrow m(E-A_\varepsilon)<\varepsilon$. $\square$.

Thm I.4.2 (Lusin) f is measurable and finite-valued on E, $m(E)<\infty$.

Given $\varepsilon>0$. $\exists F_\varepsilon$ closed $\subseteq E$ s.t. $m(E-F_\varepsilon)<\varepsilon$., $f|_{F_\varepsilon}$ continuous

Proof: Let $\{f_n\}_{n\geq 1}$ setep functions s.t. $f_n \to f$ a.e.

$\exists E_n$. s.t. $m(E_n)<\frac{1}{2^n}$. f_n is continuous outside E_n

By Egorov's thm, $\exists A_{\frac{\varepsilon}{3}}$, s.t. $f_n \to f$ uniformly in A_ε. and $m(E-A_{\frac{\varepsilon}{3}})<\frac{\varepsilon}{3}$.

Then consider $F' = A_{\frac{\varepsilon}{3}} - \bigcup_{n\geq N} E_n$ s.t. $\sum_{n\geq N} \frac{1}{2^n} < \frac{\varepsilon}{3}$.

$\forall n \geq N$. f_n is continuous on $F' \Rightarrow f$ is continuous on F'.

Choose $F_\varepsilon \subseteq F'$ closed. s.t. $m(F'-F_\varepsilon)<\frac{\varepsilon}{3}$.

Then $m(E-F_\varepsilon) = m(E-A_{\frac{\varepsilon}{3}}) + m(\bigcup_{n\geq N} E_n) + m(F'-F_\varepsilon)$.

$(E \setminus F_\varepsilon = (E-A_{\frac{\varepsilon}{3}}) \cup (\bigcup_{n\geq N} E_n) \cup (F'-F_\varepsilon))$.

$< \frac{\varepsilon}{3}+\frac{\varepsilon}{3}+\frac{\varepsilon}{3} < \varepsilon \Rightarrow f|_{F_\varepsilon}$ continuous $\square$

5. The estimation of $m(A+B)$.

Thm I.5.1. Suppose that $A, B, A+B$ measurable in $\mathbb{R}^d$. then $m(A+B)^{\frac{1}{d}} \geq m(A)^{\frac{1}{d}} + m(B)^{\frac{1}{d}}$

Proof: 1) When A, B are rectangles with lengths $\{a_j\}$, $\{b_j\}$.

Then $\left(\prod_{j=1}^{d}(a_j+b_j)\right)^{\frac{1}{d}} \geq \left(\prod_{j=1}^{d} a_j\right)^{\frac{1}{d}} + \left(\prod_{j=1}^{d} b_j\right)^{\frac{1}{d}}$

Indeed, W.L.O.G. Assume that $a_j + b_j = 1$. Then by A-G inequality we can prove it.

2) When A, B are the union of finitely many rectangles which are almost disjoint.

By translation, we can make A_+ and A_- contain at least one less rectangle than A, where $A_+ = A \cap \{x_j \geq 0\}$ where j is fixed $A_- = A \cap \{x_j < 0\}$.

Translate B s.t. $\frac{m(B_\pm)}{m(B)} = \frac{m(A_\pm)}{m(A)}$

$A+B \supseteq (A_+ + B_+) \cup (A_- + B_-)$.

By making induction on the numbers of rectangles of A & B.

$$\begin{aligned}
\Rightarrow m(A+B) &\geq m(A_+ + B_+) + m(A_- + B_-) \\
&\geq \left(m(A_+)^{\frac{1}{d}} + m(B_+)^{\frac{1}{d}}\right)^d + \left(m(A_-)^{\frac{1}{d}} + m(B_-)^{\frac{1}{d}}\right)^d \\
&= m(A_+)\left(1 + \left(\frac{m(B)}{m(A)}\right)^{\frac{1}{d}}\right)^d + m(A_-)\left(1 + \left(\frac{m(B)}{m(A)}\right)^{\frac{1}{d}}\right)^d \\
&= \left(m(A)^{\frac{1}{d}} + m(B)^{\frac{1}{d}}\right)^d
\end{aligned}$$

3) $m(A) < \infty$, $m(B) < \infty$, A, B be open.

Then $\forall \varepsilon > 0$. $\exists A_\varepsilon, B_\varepsilon$ almost disjoint rectangles s.t. $\begin{cases} m(A) < m(A_\varepsilon) + \varepsilon \\ m(B) < m(B_\varepsilon) + \varepsilon \end{cases}$

$\Rightarrow \left(m(A_\varepsilon)^{\frac{1}{d}} + m(B_\varepsilon)^{\frac{1}{d}}\right)^d \leq m(A_\varepsilon + B_\varepsilon) \leq m(A+B)$

let $\varepsilon \to 0$ we have the desired inequality.

4). A, B be compact. Then define $A^\varepsilon = \{x \mid d(x, A) < \varepsilon\}$

$A^\varepsilon \searrow A$ $(\varepsilon \to 0^+)$ $\Rightarrow A+B \subseteq A^\varepsilon + B^\varepsilon \subseteq (A+B)^{2\varepsilon}$

$\Rightarrow m(A+B)^{2\varepsilon}$ By $m(A^\varepsilon + B^\varepsilon) \geq \left(m(A^\varepsilon)^{\frac{1}{d}} + m(B^\varepsilon)^{\frac{1}{d}}\right)^d$.

$\Rightarrow m(A+B)^{2\varepsilon} \geq \left(m(A^\varepsilon)^{\frac{1}{d}} + m(B^\varepsilon)^{\frac{1}{d}}\right)^d$. let $\varepsilon \to 0^+$ we have $m(A+B) \geq \left(m(A)^{\frac{1}{d}} + m(B)^{\frac{1}{d}}\right)^d$ ■

5) General case: $\exists$ compact A_ε. B_ε. s.t. $m(A - A_\varepsilon) < \varepsilon$. $m(B - B_\varepsilon) < \varepsilon$.

Then $A_\varepsilon + B_\varepsilon$ is compact, $A_\varepsilon + B_\varepsilon \subseteq A+B$.

$\Rightarrow m(A+B) \geq \left(m(A_\varepsilon)^{\frac{1}{d}} + m(B_\varepsilon)^{\frac{1}{d}}\right)^d$ let $\varepsilon \to 0^+$ We have the desired inequality □

Chap 2. Ⅱ. Integration Theory.

1. Lebesgue Integral

Def Ⅱ.1.1. The Canonical form: of a simple function $\varphi := \sum_{k=1}^{M} a_k \chi_{E_k}$

$a_k \neq a_j$ $(\forall k \neq j)$ are all values φ could take. E_k are disjoint

Def Ⅱ.1.2. Let φ be a simple function. the Lebesgue integral of $\varphi := \sum_{k=1}^{M} c_k m(E_k)$.

Denoted by $\int_{\mathbb{R}^d} \varphi(x)\,dx$.

If E is a measurable subset of $\mathbb{R}^d$. then $\int_E \varphi(x)\,dx := \int_{\mathbb{R}^d} \varphi(x)\,\chi_E(x)\,dx$

Prop. Ⅱ.1.3. φ simple

(1) $\varphi = \sum_{k=1}^{N} a_k \chi_{E_k}$ be any representation.

Then $\int \varphi = \sum_{k=1}^{N} a_k\, m(E_k)$

(2) $\int (a\varphi + b\psi) = a\int\varphi + b\int\psi$

(3) E, F disjoint. $m(E), m(F)$ finite, then $\int_{E\cup F} \varphi = \int_E \varphi + \int_F \psi$.

(4) $\varphi \leqslant \psi$ then $\int\varphi \leqslant \int\psi$

(5) φ is simple, then $|\varphi|$ is simple, and $\left|\int\varphi\right| \leqslant \int|\varphi|$

Proof: Only (1),

Let $\varphi = \sum_{k=1}^{N} a_k \chi_{E_k}$. First assume that E_k are disjoint.

For each distinct non-zero value a among $\{a_k\}$, define $E_a' = \bigcup E_k$.

where the union is taken over k. such that $a_k = a$. then $m(E_a') = \sum m(E_k)$

Obviously, $E_a' \cap E_b' = \phi$. $(a \neq b)$. $\Rightarrow \int\varphi = \sum a\, m(E_a') = \sum_{k=1}^{N} a_k\, m(E_k)$

Now let $\varphi = \sum_{k=1}^{N} a_k \chi_{E_k}$ Consider $E_j^* = E_1' \cap E_2' \cdots \cap E_N'$

E_k' is either E_k or E_k^c. then $E_k = \bigcup E_j^*$ $(E_k \supseteq E_j^*)$. $\bigcup_{j=1}^{2^N-1} E_j^* = \bigcup_{k=1}^{N} E_k$

let $a_j^* = \sum a_k$. for all k s.t. $E_k \supseteq E_j^*$ then E_j^* are disjoint,

a_j^* are distinct. $\Rightarrow \int\varphi = \sum a_j^* m(E_j^*) = \sum_k \sum_{E_k \supseteq E_j^*} a_k\, m(E_j^*) = \sum a_k\, m(E_k)$. □

Def Ⅱ.1.4. f is measurable, the support set of f: $\operatorname{supp}(f) := \{x \mid f(x) \neq 0\}$

f is supported on E if $f(x) = 0$ whenever $x \notin E$.

Lemma II.1.5. f is a bounded function on a set E of finite measure. If $\{\varphi_n\}_{n\geq 1}$ is any sequence of simple functions bounded by M, supported on E and with $\varphi_n(x) \to f(x)$ a.e. x, then (i) $\lim_{n\to\infty} \int \varphi_n$ exists

(ii) If $f = 0$ a.e., then $\lim_{n\to\infty} \int \varphi_n = 0$.

proof: (i) By Egorov's thm. Given $\varepsilon > 0$, $\exists$ closed $A_\varepsilon \subseteq E$ s.t. $m(E - A_\varepsilon) < \varepsilon$, $\varphi_n \to f$ uniformly on A_ε.

Let $I_n = \int \varphi_n$. then $|I_n - I_m| \leq \int_E |\varphi_n - \varphi_m| dx$

$$= \int_{A_\varepsilon} |\varphi_n - \varphi_m| dx + \int_{E-A_\varepsilon} |\varphi_n - \varphi_m| dx$$

$$\leq 2m(E)\varepsilon + 2M\varepsilon \quad \text{when } m,n \text{ is large enough}$$

Thus $\{I_n\}$ converges.

(ii). If $f = 0$ a.e. then $|I_n| \leq m(E)\varepsilon + 2M\varepsilon \Rightarrow \lim_{n\to\infty} I_n = 0$ $\square$.

Def II.1.6. f is a bounded function supported on E, $m(E) < \infty$.

Then define $\int f(x)\,dx = \lim_{n\to\infty} \int \varphi_n(x)\,dx$.

$$\int_E f(x)\,dx = \int f(x)\,\chi_E(x)\,dx$$

Thm II.1.7. (Bounded Convergence Theorem). $\{f_n\}$ bounded by M, supported on E, $m(E) < \infty$, and $f_n(x) \to f(x)$ a.e. x $(n \to \infty)$. Then f is measurable, bounded, supported on E for a.e. x, (*) and $\int f_n \to \int f$ $(n \to \infty)$

Proof: (*) is obvious. By Egorov thm: find A_ε

$$\int |f_n - f|\,dx \leq \int_{A_\varepsilon} |f_n(x) - f(x)|\,dx + \int_{E-A_\varepsilon} |f_n(x) - f(x)|\,dx$$

$$\leq m(E)\varepsilon + 2M\,m(E - A_\varepsilon) \quad \text{for all large enough } n$$

$$\Rightarrow \int |f_n - f|\,dx = 0 \quad (n \to \infty) \Rightarrow \int f_n \to \int f \quad (n \to \infty) \quad \square$$

Rmk (1) that is $\lim_{n\to\infty} \int f_n = \int \lim_{n\to\infty} f_n$.

(2) $f \geq 0$, bounded, supported on E, $m(E) < \infty$ and $\int f = 0$. then $f = 0$ a.e. x.

Let $E_k = \{x \in E \mid f(x) > \frac{1}{k}\}$. then $\frac{1}{k}\chi_{E_k}(x) \leq f(x) \Rightarrow \frac{1}{k} m(E_k) \leq 0 \Rightarrow m(E_k) = 0$

Since $\{x \mid f(x) > 0\} = \bigcup_{n\geq 1} E_n$, we have $f = 0$ a.e. x. $\square$

Def Ⅱ.1.8. f is non-negative, measurable. define $\int f(x)dx = \sup_{g} \int g(x)dx$, where $0 \le g \le f$, g bounded, supported on a set of finite measure.

If $\sup < \infty$ then we say f is Lebesgue integrable.

Prop. Ⅱ.1.9. $f, g \ge 0$, integrable,

(i) $a, b > 0 \Rightarrow \int(af+bg) = a\int f + b\int g$

(ii) $E \cap F = \phi, \Rightarrow \int_{E\cup F} f = \int_E f + \int_F f$.

(iii) $0 \le f \le g \Rightarrow \int f \le \int g$

(iv) g is integrable, $0 \le f \le g \Rightarrow f$ is integrable

(v) f is integrable, then $f < \infty$ a.e. x.

(vi) $\int f = 0$, then $f(x) = 0$ a.e. x

proof: (i) $a = 1 = b$, if $\varphi \le f$, $\psi \le g$. φ, ψ bounded, supp on E, $m(E) < \infty$

then $\varphi + \psi \le f+g$. then $\varphi+\psi$ bounded, supp on E', $m(E') < \infty$

$\Rightarrow \int f + \int g \le \int (f+g)$.

Let η be bounded, supp on E, $m(E) < \infty$, $\eta \le f+g$.

$\eta_1 = \min\{f, \eta\}$, $\eta_2 = \eta - \eta_1$, then $\eta_1 \le f$, $\eta_2 \le g$.

$\Rightarrow \int \eta = \int(\eta_1+\eta_2) = \int \eta_1 + \int \eta_2 \le \int f + \int g$. Take superum of η

$\Rightarrow \int(f+g) \le \int f + \int g \Rightarrow \int f + \int g = \int(f+g)$ □

(Fatou)

Lemma Ⅱ.1.10. $\{f_n\} \ge 0$, measurable, If $\lim_{n\to\infty} f_n(x) = f(x)$ a.e. x. then $\int f \le \liminf_{n\to\infty} \int f_n$

proof: $\forall g$, $0 \le g \le f$. bounded, supp on E, $m(E) < \infty$, let $g_n = \min\{g, f_n\} \le f_n$

$g_n \to g$ a.e. $x \Rightarrow \int g_n \to \int g$ (By bounded convergence thm)

$\int g_n \le \int f_n$. Take the liminf, we have $\int g \le \liminf_{n\to\infty} \int f_n$.

Take the superum of g, we have $\int f \le \liminf_{n\to\infty} \int f_n$

Def Ⅱ.1.11. $f_n \nearrow f$ refers to $f_n \le f_{n+1}$ a.e. x, all $n \ge 1$. and $\lim_{n\to\infty} f_n(x) = f(x)$ a.e. x.

Similarly, $f_n \searrow f$.

Coro Ⅱ.1.12. (Monotone convergence theorem) $f_n \ge 0$, $f_n \nearrow f$. Then $\lim_{n\to\infty} \int f_n = \int f$.

proof: $f_n \le f$. $\Rightarrow \int f_n \le \int f$ $\forall n \ge 1$. $\Rightarrow \lim_{n\to\infty} \sup \int f_n \le \int f$.

By Fatou's thm, we immediately have $\lim_{n\to\infty} \int f_n = \int f$. □.

Coro II.1.13. $\sum_{k=1}^{\infty} a_k(x)$ where $a_k(x) \geq 0$ measurable. Then $\int \sum_{k=1}^{\infty} a_k(x)\,dx = \sum_{k=1}^{\infty} \int a_k(x)\,dx$

If $\sum_{k=1}^{\infty} \int a_k(x)\,dx$ is finite, then $\sum_{k=1}^{\infty} a_k(x)$ converges a.e. x.

The general case:

Def II.1.14. f is Lebesgue integrable if $|f|$ is integrable

Let $f^+ = \max\{f, 0\}$. $f^- = \max\{-f, 0\}$ $\Rightarrow$ $f = f^+ - f^-$.

Define $\int f = \int f^+ - \int f^-$.

Prop. II.1.15. The integral of Lebesgue integrable functions is linear, additive, monotonic, and satisfies the triangle inequality.

Proof (skipped)

Prop. II.1.16. f is integrable on $\mathbb{R}^d$. Then $\forall \varepsilon > 0$.

(i) $\exists B$, $m(B) < \infty$. s.t. $\int_{B^c} |f| < \varepsilon$.

(ii) $\exists \delta > 0$. s.t. $\int_E |f| < \varepsilon$. whenever $m(E) < \delta$.

Proof: W.L.O.G $f \geq 0$. (i) Let $f_N(x) = f(x)\,\chi_{B_N}(x)$. B_N is the ball of radius N at the origin

Then $f_N \nearrow f$. By monotone convergence theorem: $\lim_{N\to\infty} \int f_N = \int f$.

$\Rightarrow \exists N > 0$. s.t. $0 < \int f - \int f \chi_{B_N} < \varepsilon$ $\Rightarrow$ $0 < \int f \chi_{B_N^c} < \varepsilon$ $\Rightarrow$ $\int_{B_N^c} f < \varepsilon$

(ii). $\forall \varepsilon > 0$. $\exists N > 0$. s.t. $0 < \int f - \int f_N < \frac{\varepsilon}{2}$. Choose $\delta > 0$. s.t. $N\delta < \frac{\varepsilon}{2}$. $\forall m(E) < \delta$.

$$\int_E f = \int_E (f - f_N) + \int_E f_N$$

$$\leq \int (f - f_N) + N m(E) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \square$$

Thm II.1.17. (Lebesgue dominated convergence theorem) f_n measurable, $f_n(x) \to f(x)$ a.e. x.

If $|f_n(x)| \leq g(x)$, $g(x)$ is integrable. then $\int f_n \to \int f$ $(n \to \infty)$

Proof: $\forall N \geq 0$. 令 $E_N := \{x \mid |x| \leq N,\ g(x) \leq N\}$, $\exists$ large N. s.t. $\int_{E_N^c} g < \varepsilon$.

$\Rightarrow \int_{E_N} f_n$ is bounded, supp on E, $m(E) < \infty$

By bounded convergence thm, $\int_{E_N} |f_n - f| < \varepsilon$. $\forall n$ large

$\Rightarrow \int |f_n - f| \leq \int_{E_N} |f_n - f| + 2\int_{E_N^c} g \leq 3\varepsilon$.

Ⅲ. Abstract Integration. 1. Abstract measures

Def Ⅲ.1.1. $\mathfrak{R} \subseteq \mathcal{P}(X)$ is a σ-Algebra in X if:

(i) $X \in \mathfrak{R}$

(ii) If $A \in \mathfrak{R}$, then $A^c \in \mathfrak{R}$

(iii) If $A = \bigcup_{n=1}^{\infty} A_n$. $A_n \in \mathfrak{R}$. then $A \in \mathfrak{R}$

Then X is called a measurable space. Elements of $\mathfrak{R}$ are measurable sets

Def Ⅲ.1.2. $f: X \to Y$. X measurable space. Y topological space

f is measurable iff $f^{-1}(V)$ is measurable all open sets V.

Let X be a measurable space.

Def Ⅲ.1.3. A measure $\mu: \mathfrak{R} \longrightarrow [0, +\infty]$. with the following property:

$E_i \in \mathfrak{R}$. Then $\mu(\bigcup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} \mu(E_n)$.

Def Ⅲ.1.4. Let X be aset, An exterior measure $\mu_* : \mathcal{P}(X) \longrightarrow [0, +\infty]$.

that satisfies: (i) $\mu_*(\phi) = 0$

(ii) If $E_1 \subseteq E_2$. then $\mu_*(E_1) \leq \mu_*(E_2)$.

(iii) $\forall \{E_k\}_{k \geq 1}$. we have $\mu_*(\bigcup_{k \geq 1} E_k) \leq \sum_{k \geq 1} \mu_*(E_k)$

Def Ⅲ.1.5. A set E in X is measurable iff $\mu_*(A) = \mu_*(E \cap A) + \mu_*(E^c \cap A)$. $\forall A \subseteq X$.

Thm Ⅲ.1.6 (Carathéodory theorem) Given an exterior measure μ_* on a set X, the collection $\mathfrak{R}$ of measurable sets forms a σ-algebra. Moreover, $\mu_*|_{\mathfrak{R}}$ is a measure

Proof:

2. Metric exterior measures. (Now we assume that X is a metric space).

Def III.2.1. X is a metric space. The Borel σ-algebra $\mathcal{B}_X$ is the smallest σ-algebra of sets X that contains the open sets in X. Elements in $\mathcal{B}_X$ are called Borel sets.

Def III.2.2. An exterior measure μ_* on X is a metric exterior measure if it satisfies

$$\mu_*(A \cup B) = \mu_*(A) + \mu_*(B) \quad \text{whenever } d(A,B) > 0.$$

Thm III.2.3 μ_* is a metric exterior measure on ~~a set~~ X, then the Borel sets in X are measurable. Hence $\mu_*|_{\mathcal{B}_X}$ is a measure.

Proof:

Given a metric space X, μ defined on the Borel sets of X is called the Borel measure.

Prop Ⅲ.2.4. Suppose the Borel measure μ is finite on all balls in X of finite radius. Then for any Borel set E, $\forall \varepsilon > 0$. $\exists$ O open. F closed. s.t. $F \subseteq E \subseteq O$ $\mu(O - E) < \varepsilon$, $\mu(E - F) < \varepsilon$

Proof: